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Emission Lines Signatures of Runaway Supermassive Black Holes

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ABSTRACT

The orbiting of a supermassive black hole binary system is a massively energetic phenomenon that can have significant dynamical effects within active galactic nuclei (AGN). Eventually, these two supermassive black holes (SMBHs) may spiral into each other and coalesce into a single larger SMBH. As a result of this collision and the emission of gravitational waves from the orbiting bodies, the merged black hole can receive a 'kick' velocity of up to several 1000 km/s, enough to eject it from the center of its host galaxy. As the black hole leaves the galaxy's center, it drags some of the surrounding gas with it, changing the broad emission lines we normally expect to observe in an AGN and allowing us to detect the recoiling SMBH. In this project, I used an N-Body code to simulate the effect of a SMBH recoil on the orbits of the surrounding gas clouds. The simulation output generates the position and velocity vectors of the clouds from which we can calculate the emission line profiles of the system as it evolves. Further analysis of the parameter space gives us a set of characteristics that can be used as guides when searching for recoil candidates in spectroscopic databases.

1. INTRODUCTION

1.1. *Black Holes*

Since their discovery in the early 1970s, black holes have been a popular subject of study in astronomy. These enigmatic objects form from the collapse of massive stars and possess gravitational forces so intense that they warp spacetime itself to such an extreme extent that not even light can escape. However, supermassive black holes must grow from these lower-mass 'seeds' either by accreting gas or merging with other black holes.

Although black holes have fascinated scientists and the general public for decades, numerous fundamental questions about their nature, behavior, and evolution remain unanswered. This capstone project seeks to discover more about supermassive black hole binaries and how they can merge and evolve. By creating theoretical models and comparing them with characteristics of observed spectra from active galactic nuclei, I aim to assist in the search for these rare recoil events.

1.2. *Supermassive Black Hole Binaries*

SMBHs are present at the center of nearly every galaxy in the universe. During periods of growth, when the SMBH accretes surrounding gas, the emission lines produced by the radiation of the SMBH due to accretion are much more prominent than those of a normal galaxy. We call these active galactic nuclei (AGNs). The energy released during this period strongly influences the overall life cycle of a galaxy.

We believe that SMBH binaries form during galaxy mergers due to dynamical interactions with stars and the two SMBHs will sink towards the center of mass of the merged galaxy and form a binary. These binaries can remain in a stable state for a very long time, however, their orbits will eventually decay quickly because of the emission of gravitational waves, causing them to coalesce into a single larger SMBH.

This research project focuses on the aftermath of merging of the binary SMBHs. Overall, the goal of this project is to characterize the emission line signatures resulting from the high-speed recoils of the merged SMBHs.

As a result of conservation of linear momentum, as well as the orientation of the SMBH spins relative to each other, and the fact that the gravitational wave emission is not uniform in direction, the merged SMBH can recoil and receive a 'kick' velocity of several 1000 km/s (Gualandris & Merritt 2008).

This results in the displacement of the SMBH from the center of the galaxy. If the kick velocity is large enough, it could completely eject the SMBH from its host. As the SMBH is ejected, it can carry its accretion disk and surrounding gas clouds, also known as the broad-line region, along with it. Whether or not a cloud will be carried along is based on its Keplerian velocity at the time of the recoil. If the Keplerian velocity is greater than the magnitude of the 'kick' velocity, it is much more likely to follow along than a cloud with a Keplerian velocity less than the magnitude of the kick. The displacement of these clouds will appear as an offset AGN and is the main characteristic we look for when identifying recoils.

1.3. *Broad and Narrow-line Region*

In normal AGN, the SMBH and its accretion disk are surrounded by a dusty torus structure mainly made up of molecular gas and dust clouds. This dust usually sublimates at temperatures between 1500-1800 K, which means it cannot survive close to the accretion disk of the SMBH. The radius at which this dust can no longer exist is referred to as the dust sublimation radius (Rd). The area inside the dust sublimation radius, populated with dense gas clouds that are photoionized by the radiation of the accretion disk (Baskin & Laor 2018), is the broad-line region (BLR). The gas that exists in this region is very strongly gravitationally bound to the SMBH and emits Doppler-broadened emission lines that are broadened by several thousand km/s (Netzer 2008). Although the BLR is too small to spatially resolve, its unique spectroscopic properties allow us to use it as a unique tool to obtain more information about the SMBHs at the center of AGN.

Just outside this region, further away from the dusty torus and accretion disk, is the narrow-line region (NLR). This region forms a cone shape that emerges from either side of the torus. This region also contains photoionized gas clouds, illuminated by the radiation escaping from the SMBH at the center of the torus. However, these clouds are not as strongly gravitationally bound to the SMBH as the BLR, meaning they are likely to be left behind in the presence of a SMBH recoil. As the name suggests, the emission lines produced by this region, while still broad, are much narrower than those of the BLR in comparison.

1.4. *Spectroscopic Analysis*

In spectroscopic surveys, recoiling AGN are found by detecting the Doppler shift of the broad emission line caused by the motion of the SMBH; looking for these Doppler broadened emission lines is what we will use to detect the displacement of the SMBH from its AGN (Bogdanović et al. 2022). When looking at the emission line profiles, as the SMBH and BLR move away from the regions of the AGN, it appears to "leave behind" the emission lines produced by the NLR and pull the BLR emission lines with it. The narrow emission lines would not 'follow' a recoiling SMBH, and the broad emission lines would become displaced due to the Doppler shift as it moves away from the galactic center. An example of this is given in Figure 1.

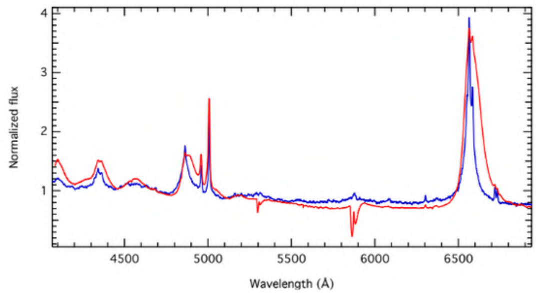


Figure 1: The spectrum of the quasar E1821+643, a gravitational recoil candidate, in red (Jadhav et al. 2021) with an average quasar spectrum, in blue (Berk 2001), to compare. Note, the peaks of the broad lines are shifted to longer wavelengths.

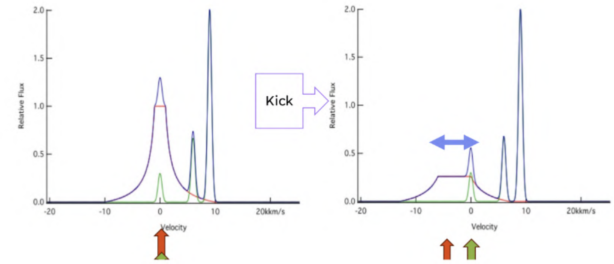


Figure 2: This figure was created by Dr Robinson to show the expected velocity shift and change in the FWHM in the presence of a kick velocity. The arrows indicate the shift of the broad emission lines from the narrow emission lines.

Following the recoil of the SMBH, the broad line is both Doppler shifted relative to the narrow lines and may become broadened further as the slower-moving gas clouds in the outer edges of the broad line region are left behind. This broadening is illustrated schematically in Figure 2.

1.5. *Rebound and the AGN-Body Code*

Rebound (Rein & Liu 2012) is an open-source Python N-body integrator with many applications in material physics and astrophysics. In this project, we used Rebound to create an N-body simulation of a recoiling SMBH and its dynamical effects on the surrounding broad-line region. Because the gas clouds that make up the broad-line region have negligible mass when compared to the mass of the SMBH, in this simulation, an individual gas cloud is represented as a massless test particle. Treating the individual particles as massless enables us to speed up the run time of the simulation quite a bit while still providing an overall accurate representation of the system.

In our simulation, we specifically use the Integrator with Adaptive Step-size control, 15th order (IAS15) (Rein & Spiegel 2015). IAS15 is a high-order non-symplectic integrator primarily used to handle velocity-dependent arbitrary forces and can automatically choose timesteps as the simulation integrates. While this algorithm for choosing timesteps is useful, it means that we cannot control the value of the timestep. Instead, we can only set a minimum allowed timestep for the simulation to use as it integrates. If we wanted to manually set the timestep and raise it by a factor of 10, we would end up increasing the error by a factor of 10^{15}. Still, providing Rebound with a minimum timestep prevents things from getting too slow and is sufficient for this simulation.

During the summer of 2024, I spent 10 weeks creating the first simulation that models the physical effects of a SMBH recoil on the BLR of an AGN using an N-body code. The goal during these 10 weeks was to create a program that I would then use and refine in my capstone project starting fall 2024. In this code, we first initialize the system by providing it with pa-

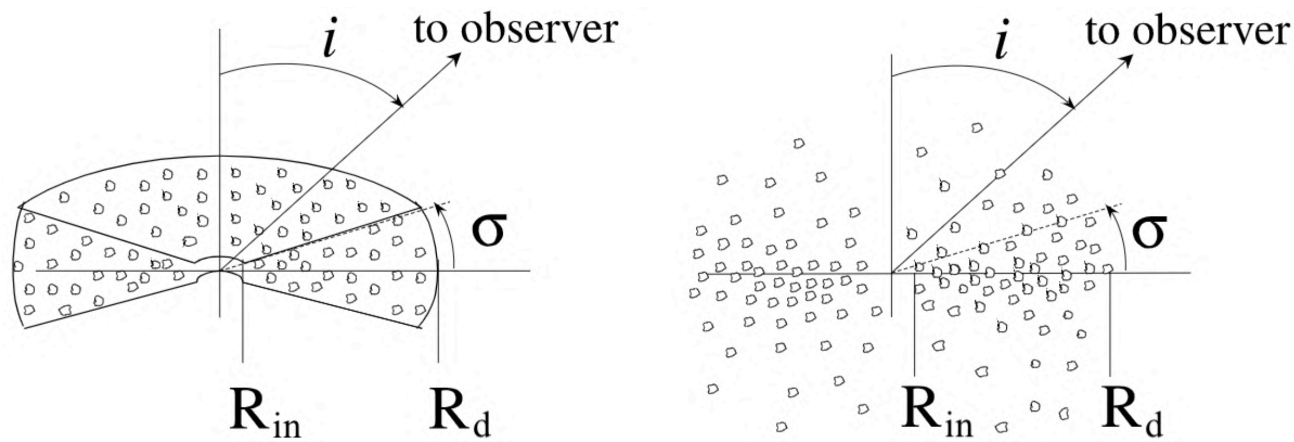


Figure 3: The expected structure of the BLR (Rosborough et al. 2023). The left shows a wedge shape that arranges the clouds in a flared disk. R_{in} and R_d are the inner and outer radii, σ represents the half-angle width of the disk, and i is the inclination of the disk axis to the line of sight (LOS) of the observer. On the left is a modified version of the first wedge that blurs the edges to make it more representative of what we would expect the actual structure to be. The left structure is what is used in the AGN-Body Code simulations.

rameters such as black hole mass, kick velocity, the number of gas clouds, and the number of timesteps we want the simulation to run for, in addition to the range of space the particles can initially exist in. From there, we provide the central mass with a kick velocity and let the simulation run to see how the BLR reacts.

2. METHODS

2.1. *Luminosity Calculations*

For my Capstone I project, I created and developed the Active Galactic Nuclei- Broad-line Emission And N-body Simulation (AGN-BEANS) code. This combines the N-body integrator Python package Rebound as well as a BLR model based on basic photoionization physics. That allows for the simulation and generation of spectra modeling SMBHs and their surrounding broad-line region. Using the code described in section 1.5 as a base, we first extract the position and velocity vectors corresponding to each timestep in the N-body orbital simulation and combine them into a database.

Variable	Range
Inclination	$-0.1-+0.1$
Longitude of ascending node	$0-\pi$
True anomaly	$([0-1])*2\pi$

Table 1: This table shows the ranges of three initial orbital variables that are randomized for each particle in the simulation. These values determine the initial shape of the cloud distribution and are represented in radians.

The initial configuration of the particles is modeled as a flared disk, with the clouds orbiting the SMBH at a range of inclinations. Figure 3 shows an illustration of the expected structure. To create this configuration, we randomly assign each particle's orbital elements within certain limits. The specific parameters and corresponding values are detailed in Table 1.

After providing the code with the timesteps you wish to see the profile for, it will retrieve the position and velocity vectors for the given number of particles. From there, it will calculate the line of sight (LoS) velocity based on the desired inclination of the observer, also provided by the user.

This inclination can be any value between 0 and 90 degrees. An inclination of 90 degrees would indicate you are looking at the disk edge-on, and the LoS velocity is given by the y-velocity of each particle. An inclination of 0 degrees represents the observer viewing the full disk along the rotation axis, with the z-velocity used as the LoS velocity instead. For any value not equal to 90 or 0 degrees, a rotational transformation about the x-axis adjusts the values so that the transformed z-velocity will represent the new LoS velocity.

Once the code calculates the LoS velocity, each particle is assigned a set of parameters needed to calculate its luminosity. These parameters are shown in Table 1.

One of the variables in this table is the ionizing photon flux incident on the cloud, which is defined by;

$$F(r) = \frac{Q}{4\pi r^2} \tag{1}$$

In this equation, Q represents the ionizing photon luminosity of the accretion disk, and r is the distance of the cloud from the SMBH, computed from the cloud's position at each timestep. The ionizing photon luminosity is scaled to the luminosity of the AGN itself, which is derived from the Eddington ratio and the bolometric luminosity of the galaxy. This parameter represents the ratio of the ionizing photon flux to the gas density at distance r.

Another variable is the ionization parameter, represented by;

$$U(r) = \frac{F(r)}{cn(r)} \tag{2}$$

After these parameters are assigned, we use these values to determine the Strömgren depth (d_s) and the cloud depth (d_c). The value of d_c is proportional to the average path length through

Description	Variable	Values
Cloud radius	R_{cl}	$10^{12} - 10^{14}$ cm
Cloud density	n(r)	$10^8 - 10^9$ $\mathrm{H-atoms/cm^3}$
Ionizing photon flux	F(r)	Used to calculate U(r)
Ionization parameter	U(r)	Calculated from F(r) and n(r)

Table 2: Similar to the initial orbital parameters, the cloud radius and density are also assigned randomly from the provided range. These ranges are based on values thought to be standard to the BLR.

the cloud, which is, in turn, proportional to R_{cl}. We calculate d_s, which represents the depth to which the clouds are ionized by the incident radiation from the accretion disk, using the following equation;

$$d_s = \frac{cU(r)}{n(r)\alpha_B} \tag{3}$$

Where α_B represents the recombination coefficient for hydrogen ($\alpha_B \approx 2.6 \times 10^{-13}$ cm^{3}s^{-1}).

By calculating these two values, we can determine whether the cloud is fully ionized or partially ionized. If $d_s \geq d_c$ the cloud is fully ionized and is "matter-bounded"; if $d_s < d_c$ it is considered "radiation-bounded" and only partially ionized. Whether a gas cloud is considered matter-bounded or radiation-bounded changes how we calculate the luminosity of that specific cloud, shown in the following equation;

$$L_{cl} = \begin{cases} n(r)^2 \alpha_{H\alpha}^{eff} h\nu_{H\alpha} \pi R_{cl}^2(r) d_s(r), & \text{for } d_s < d_c \\ n(r)^2 \alpha_{H\alpha}^{eff} h\nu_{H\alpha} \pi R_{cl}^2(r) d_c(r), & \text{for } d_s \geq d_c \end{cases} \tag{4}$$

Where $\alpha_{H\alpha}^{eff}$ is the effective recombination coefficient for h-alpha emission and $h\nu$ is photon energy of h-alpha.

From this equation, the code calculates the individual H-alpha cloud luminosities for each cloud, adding them together and binning them in LoS velocity to get the total H-alpha line luminosity to create the line profiles described in the next section.

2.2. *Generating Spectra*

Once AGN-BEANS has calculated and stored the luminosity and LoS velocity of each particle, it then begins the process of generating the spectrum for the given timestep. First, the particles are sorted by LoS velocity and divided into velocity bins of around 100 km/s in width. Then, using the SciPy statistical function *binned_statistic*, it takes the sum of the luminosities of each particle in each bin and plots the spectrum. An example of the spectrum output of AGN-BEANS is shown in Figure 4.

3. CAPSTONE I WORK

3.1. *Results*

This section details the initial tests of AGN-BEANS to verify that it produces reasonable cloud distributions and line profiles.

TEST 1

For this test, we compare the spatial distribution of the clouds and corresponding spectra of different timesteps in a single simulation while keeping all initial variables constant. There are 4 snapshots used for this comparison, one at t = 0 and three more at roughly 20, 100, and 200 years after the kick. The kick velocity used for this test is 1000 km/s, and the simulation is considered from a 0 and 90 degrees observer inclination. The other fixed parameter values in the simulation are as follows.

- SMBH Mass: 6.5×10^{8} Solar Masses
- Number of Particles: 10,000
- Inner Radius of BLR: 6.172×10^{11} km
- Outer Radius of BLR: 1.234×10^{13} km

The full results for test 1 are detailed in Figure 5.

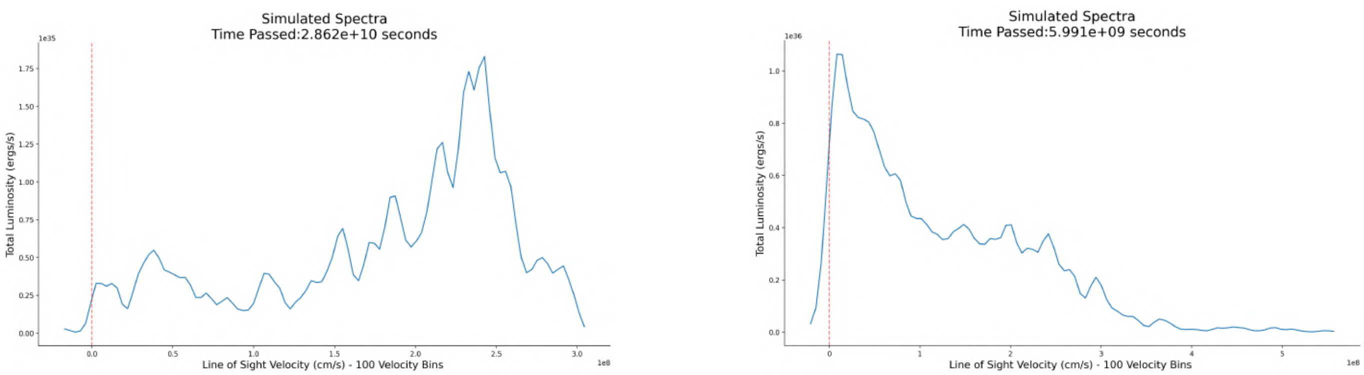
The goal of test 1 is to display how the overall evolution of the system progresses as time goes on for a kick velocity of 1000 km/s from an observer inclination of 0 and 90 degrees.

The results for the spatial distribution of the 0-degree inclination, shown in the second timestep, show that the central clouds that are strongly gravitationally bound have begun to move with the SMBH. We see this reflected in the spectra as well. It now has an extended wing due to the clouds that begin to follow the SMBH. For 90 degrees, the profile is not shifted when observing it edge-on as opposed to the shift we see for an observer angle of 0 degrees.

After 100 years, we now start to see rings forming in the distribution of the clouds that alternate due to the clouds continually falling behind and attempting to catch up. The emission profile has also been broadened quite a bit by the collection of particles keeping up with the SMBH. It also shows a large shift relative to v = 0. In the cloud distribution for 90 degrees, we see a cone shape that is consistent between all kick velocities below 3000 km/s. The 'M' shape in the profile is also broadened by the clouds closest to the SMBH.

200 years after the kick, there are now even more rings in the 0-degree perspective, as nearly all of the particles have begun to follow the SMBH. The corresponding spectrum has also made another drastic shift. Those bright and fastest-moving, most central particles are now the main contributors to the emission line profile, producing a strong peak around the kick velocity. The shape present in the spectrum for 90 degrees, however, is much more subtle. The small peaks on either side of the previous timestep have now changed to a slight broadening of the 'M' shape.

Overall, this test behaves as we expected. We see that the subsection of the particles that are closely bound to the SMBH initially remain bound after the kick, and the luminosity in the profiles reflects this. The condition for whether



(a) Example 1: $5 \times 10^7 M_\odot$ - 2,000 km/s - 0°

(b) Example 2: $5 \times 10^7 M_\odot$ - 2,000 km/s - 45°

Figure 4: Two examples of outputs generated by AGN-BEANS representing two separate simulations using a SMBH mass of $5 \times 10^7 M_\odot$, a kick velocity of 2,000 km/s, and a kick angle of 0° and 45° respectively. The red dashed line represents a LoS velocity of 0 km/s.

a particle is likely to remain bound is given by the following equation;

$$v_{Kepler} = \sqrt{\frac{GM}{r}} > v_{kick} \tag{5}$$

For the parameters in this simulation, the lowest Keplerian velocity is approximately 2600 km/s, which is much higher than the kick velocity of 1000 km/s. The farther the particle is away from the center of mass initially, the slower its Keplerian velocity will be. Which ring they become a part of in the simulation as it progresses is also dependent on the initial Keplerian velocity of the gas cloud. This idea is explored further in the Capstone II section of this paper.

TEST 2

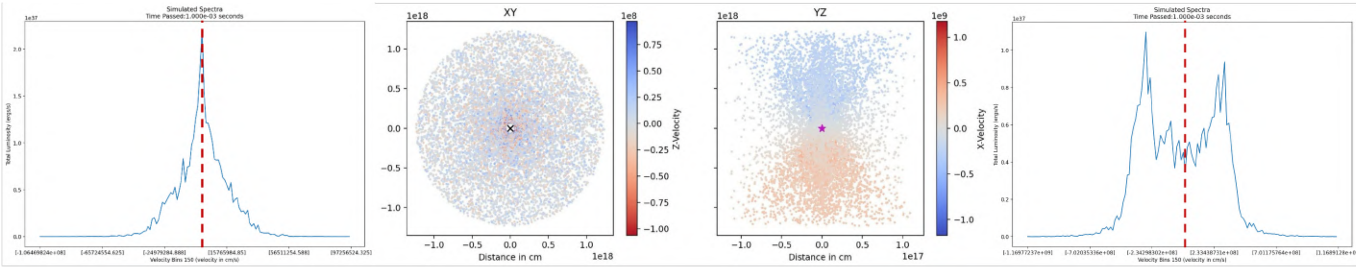
The second test compares single timesteps for two simulations with two different kick velocities (500 km/s and 2000 km/s) at a 0-degree observer inclination. Our goal is to compare how the spatial distribution and the spectra for a given time change in the presence of different kick velocities. The 4 timesteps equate to a period of 0, 20, 100, and 140 years following the kick. This test also uses the same constant parameters as Test 1, as listed previously.

The full results for the test 2 cloud distribution and spectra evolution can be found in Figures 6 and 7, respectively.

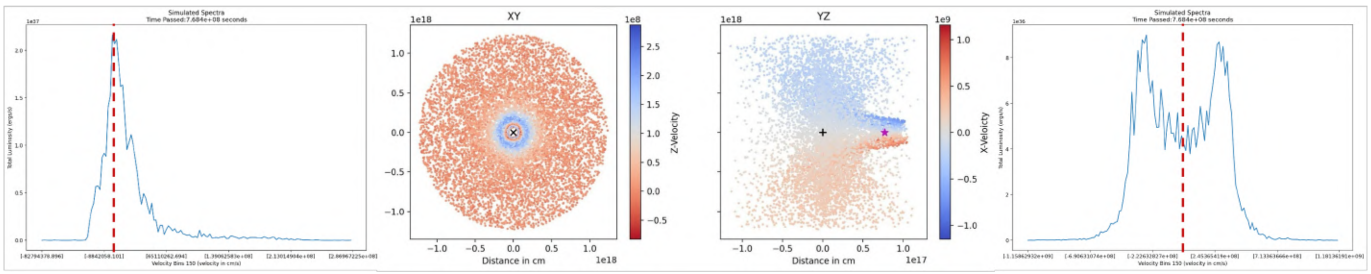
Test 2 shows the difference between the two possible kick velocities. This time, we are only viewing the system from an observer inclination of 0 degrees. As expected, the two systems evolve similarly, but the high kick velocity results in a much faster evolution.

Figure 6 shows the spatial distribution of the clouds as the system evolves. There are not many differences between the two, as both develop the same patterns, but the rings that form for the lower kick velocity have a slightly larger width. These rings show the velocity magnitude and direction, with the red clouds being left behind and the blue catching up. Using that knowledge, we also see small oscillations in the particles closest to the SMBH after fully catching up. As the SMBH continues to move away, they repeatedly lag behind before catching up, creating these characteristic ring patterns.

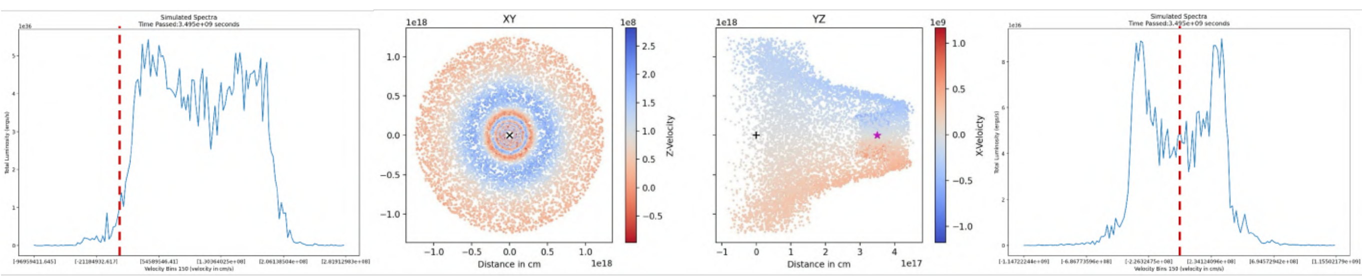
In Figure 7, the spectra show a more significant difference between the two velocities. In the second timestep, the higher kick has a much more dramatic change in the profile, however, they are both very similar in shape. The next step shows where the two profiles have diverged significantly. Even though the same amount of



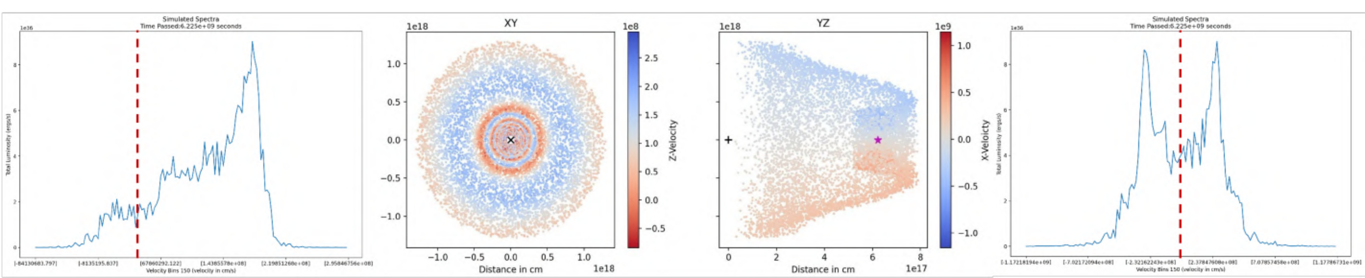
(a) t = 0 s



(b) t ≈ 25 years



(c) t ≈ 100 years



(d) t ≈ 200 years

Figure 5: Test 1: The cloud distribution and H-alpha line profiles at 4 timesteps for the simulation described in Test 1. The spectra on either side correspond to the cloud distribution next to it, with 0 degree observer inclination on the left, and 90 degrees on the right for a kick velocity of 1,000 km/s. In each image for all of the tests run during Capstone I, the color map represents the velocity scale of the individual particles; however, due to this being an early version of the code, this velocity scale —while correct—is not consistent between each timestep. This issue is fixed in the Capstone II version of the code. The black cross represents the initial position of the SMBH, the star indicates its current position, the red line in each profile image represents a LoS velocity of 0 on the velocity scale, and all units are in cgs.

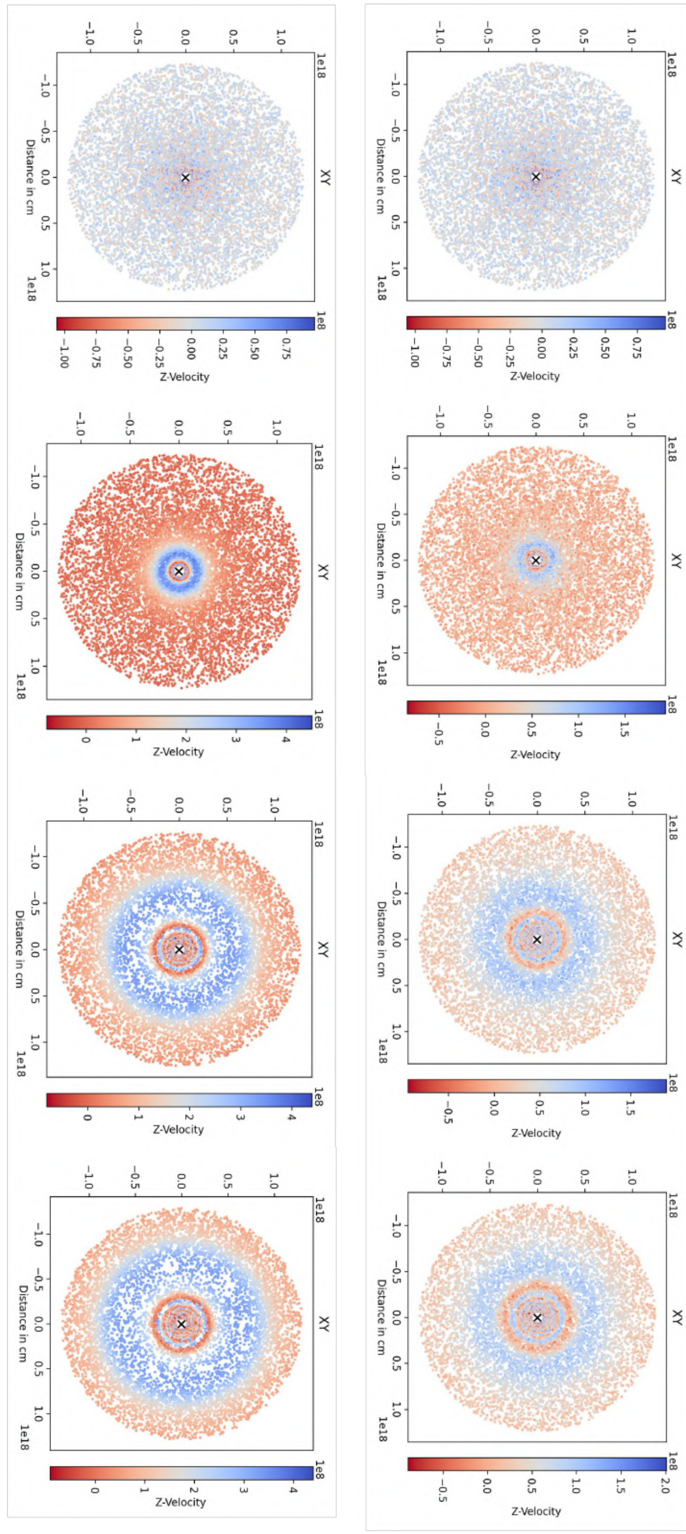


Figure 6: Test 2.1: The two columns show the evolution of the cloud distribution for a 0-degree observer inclination and a kick velocity of 2000 km/s shown on the left and 500 km/s on the right. In each image, the color bar present on the sides represents the velocity scale of each particle and varies with each snapshot. The black cross represents the initial position of the SMBH, and all units are in cgs.

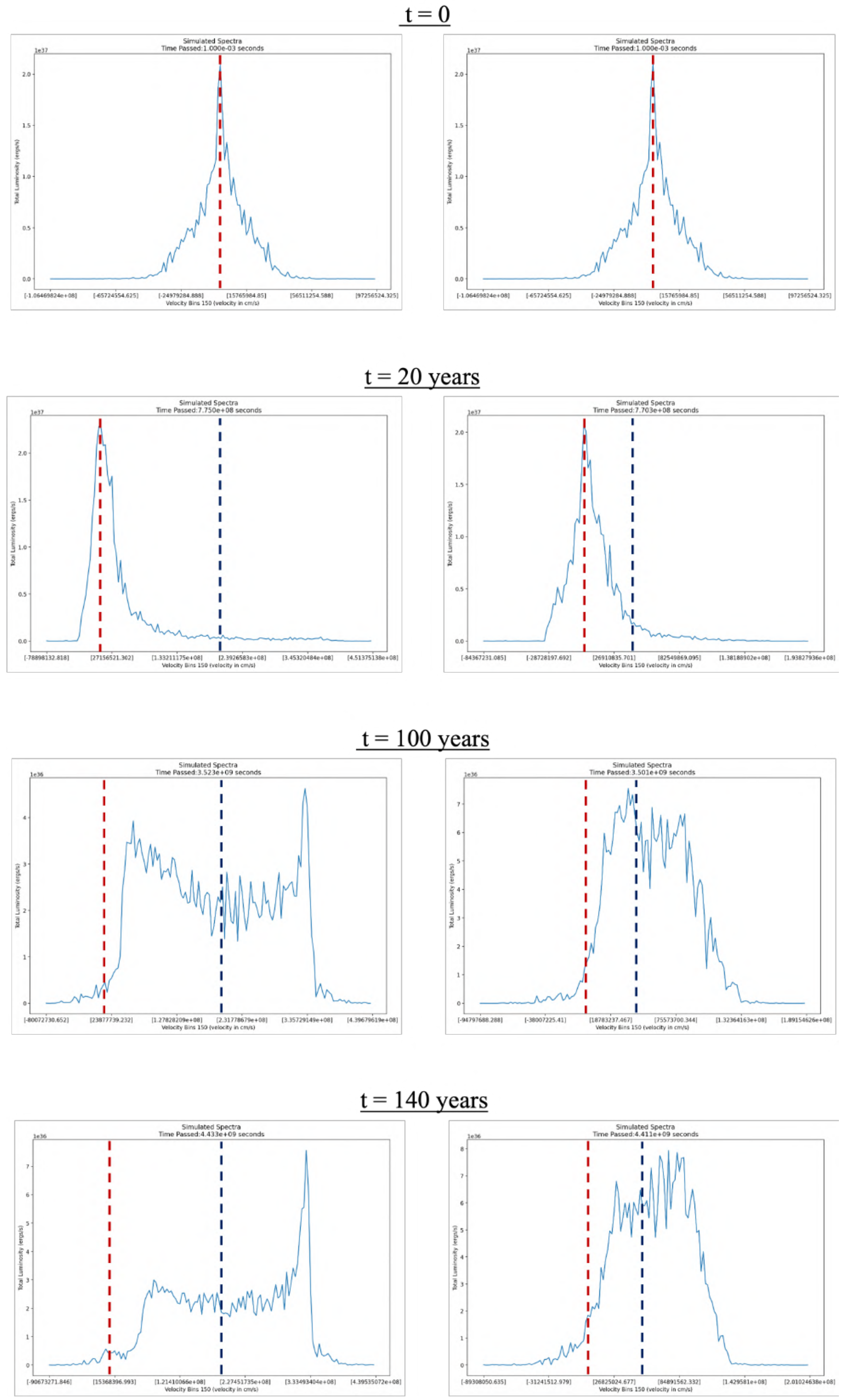


Figure 7: Test 2.2: The two columns show the evolution of the spectra for a 0-degree observer inclination and a kick velocity of 2000 km/s shown on the left and 500 km/s on the right. The red line indicates a velocity of 0, while the blue indicates the corresponding kick velocity. All units are in cgs.

time has passed in either simulation, the higher kick has already reached the point where the most central particles are starting to become the dominant contribution to the profile. The lower kick still shows the point where the entire profile is broadening. In the final timestep, we can see the peak for the higher kick has nearly doubled in luminosity, while the lower kick is still broadening.

This test is consistent with what we saw in the first test. The profiles and distributions all develop in the same steps. For the second test, we see that the only effect of a change in kick velocity is how quickly the profiles move through those steps and how many clouds remain bound to the SMBH.

4. CAPSTONE II WORK

The main goal of Capstone II was to explore the parameter space using the completed AGN-BEANS code and create a set of profile characteristics to apply to data within large databases for easier identification. The three parameters used to compare were the kick velocity, the kick angle, and the SMBH mass.

4.1. *Updates to AGN-BEANS*

One of the first steps of Capstone II was to update AGN-BEANS based on feedback from Capstone I and to implement optimizations for both the calculations and output. The main updates include;

- Vastly improving the readability of the line profile graphs.
- Added the ability for the code to calculate and scale the bolometric luminosity of the AGN based on the provided SMBH mass and Eddington ratio.
- Scaling the ionizing photon luminosity based on the new AGN luminosity.
- Application of Gaussian smoothing to reduce general noise and make the profile features more apparent.

To scale the bolometric luminosity of the AGN, the code uses the SMBH mass and the Eddington ratio provided by the user initially to calculate the Eddington luminosity. The user provides the Eddington ratio, which represents the ratio between the AGN bolometric luminosity and the Eddington luminosity.

$$\lambda \equiv \frac{L}{L_{\mathrm{Edd}}}, \tag{6}$$

where,

$$L_{\mathrm{Edd}} \equiv lM_{\mathrm{BH}} = 1.26 \times 10^{38} \left(\frac{M_{\mathrm{BH}}}{M_{\odot}}\right) \mathrm{erg\ s^{-1}} \tag{7}$$

By assuming the continuum spectrum doesn't change, the ratio $\frac{Q(M_0)}{L_{AGN}(M_0}$ remains constant as well. Because these two are constant, we can scale to other SMBH masses by using:

$$Q(M) = L_{AGN}(M)\frac{Q(M_0)}{L_{AGN}(M_0)} \tag{8}$$

Where M_0 refers to the reference black hole mass. These two simple changes mean that the simulation will properly scale based on the provided SMBH mass and Eddington ratio.

These small optimizations for the AGN luminosity and ionizing photon luminosity make the simulations more reflective of what we expect in real line profiles from databases. Another approximation that affects the simulation's accuracy is the fact that all particles in the simulation will follow the SMBH as it moves away due to there being no other mass present, as mentioned in a previous section. Addressing this would require further work that goes beyond the scope of this project, however, we do not expect this to significantly alter the main results.

4.2. *Defining the Parameter Space*

The parameter space is made up of 3 main variables: SMBH mass, kick velocity, and kick angle. This results in 27 total base simulations that I have broken down into grids to make visualization easier.

The parameters used for the kick angles and the kick velocity were chosen based on analysis of recoil simulations by Lousto et al. (2012) and Blecha et al. (2016). These two papers give us a range of probabilities for both kick angle and kick velocity, shown in Figure 8. For the kick angle described in the paper by Lousto et al. (2012) we see that the most probable angle is in the range of $10° - 15°$ this lead us to choose the values of 0°, 13°, and 45° to cover a range of angles that we can reasonable expect to see. For the kick velocity analysis done by Blecha et al. (2016), we see that the most probable velocity for AGN with various properties is around 100 km/s, however, some models predict significant probabilities for kick velocities $\geq$ 1,000 km/s that would be more likely to produce observable effects on the line profiles. For this kick velocity comparison, we decided to use the values of 1,000, 2,000, and 3,000 km/s.

4.3. *Results*

For Capstone II, once I had compiled the 27 different simulations, I did a preliminary analysis using 1,500 particles to determine which simulations would be re-run with more particles for a more in-depth analysis. I selected 9 simulations out of the original 27 based on the specific parameter values detailed previously, to run with 10,000 particles and compare the effects of kick velocity, kick angle, and SMBH mass on the system. Table 3 details the parameters for the 9 simulations used in the 3 different comparisons.

For each of the 3 initial comparisons, an observer inclination of 0 degrees was used. Following those tests, I also chose a single parameter set to see how the observers' angle relative to the AGN affects the line profile shape. For each of these tests, the comparison is made using profiles based on single snapshots long after the initial kick, when the simulation has settled into an approximately steady state.

4.3.1. *Kick Velocity Comparison*

Varied Simulation Parameters:
1,000 km/s, 2,000 km/s, 3,000 km/s

The results for this test (shown in Figure 9) are somewhat similar across all three simulations. Each one shows two distinct peaks, with the ones with LoS velocity $v_{LoS} \approx 0$ km/s representing the clouds left behind and the ones near $v_{LoS} \approx v_{kick}$ representing the clouds that remain closely bound. The main difference between the profiles comes from the large difference in the range spanned by the line of sight velocity. The profile corresponding to a kick velocity of 1,000 km/s ranges from -200 km/s 10 700 km/s in v_{LoS}. For the e 2,000 km/s and 3,000 km/s kicks v_{LoS} scales to 3,500 km/s and 4,500 km/s, respectively. There is also a large difference in luminosity, with the brightest peak in each simulation differing by an order of magnitude. For the lowest kick velocity the profile peaks at 10^{36} ergs/s and the next two peak at 10^{35} and 10^{34} ergs/s.

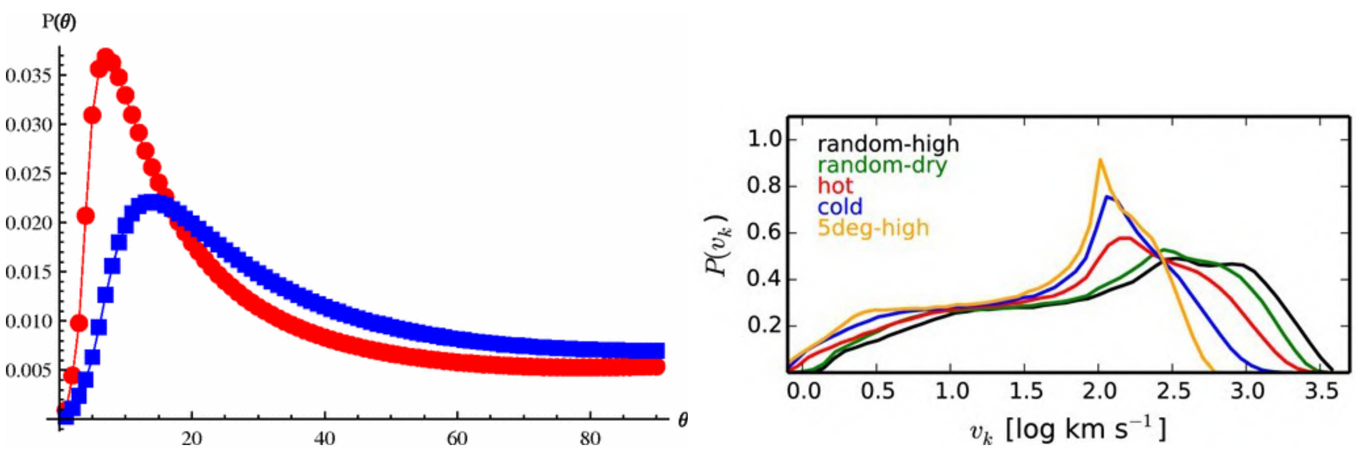
4.3.2. *Kick Angle Comparison*

Varied Simulation Parameters:
0°, 13°, 45°

For the kick angle comparison (Figure 10), we see similar profiles for the 0 and 13-degree kick angles and a very different shape for the 45-degree angle. The profiles for 0 and 13 both have a very bright peak just above the kick velocity with a dimmer peak closer to $v_{LoS} = 0$, closer to where $v = 0$. The dimmest peak for 13 degrees is twice as high as the dimmest peak for 0 degrees, but both are centered near 500 km/s. For both 0° and 13°, the brighter peak is centered close to the original kick velocity.

Description	Sim 1	Sim 2	Sim 3
Velocity Comparison $5 \times 10^7 M_\odot$, 13°	1,000 km/s 100 %	2,000 km/s 3.84%	3,000 km/s 0.13%
Angle Comparison $5 \times 10^7 M_\odot$, 2,000 km/s	0° 3.79 %	13° 3.85 %	45° 5.59%
Mass Comparison 45°, 2,000 km/s	$5 \times 10^6 M_\odot$ 0 %	$5 \times 10^7 M_\odot$ 3.85 %	$5 \times 10^8 M_\odot$ 23.61 %

Table 3: Parameter space grid showing the 9 simulations used to compare kick velocity, kick angle, and SMBH mass as well as the values for the percentage of clouds that remain bound to the SMBH in each simulation, determined by their initial Keplerian velocity.



(a) Kick Angle Distribution

(b) Kick Velocity Distribution

Figure 8: These graphs from Lousto et al. (2012) (Left) and Blecha et al. (2016) (right), respectively, show the theoretical ranges for possible kick angles and velocities. The blue and red like in the kick angle distribution graph represent hot (red) and cold (blue) environments. For the kick velocity distribution, each colored line represents different assumptions about the galaxy's gas content and the SMBH spin orientations. The selections for the parameter space were made based on these two graphs.

For the 45-degree angle, we see a triangle shape with a peak slightly offset from $v = 0$ and a plateau from 1,000 km/s to ≈2,500 km/s before it continues sloping downward towards higher v_{LoS}. Similar to the previous 2 simulations, this plateau is centered at the kick velocity.

4.3.3. *SMBH Mass Comparison*

Varied Simulation Parameters:
$5 \times 10^6 M_\odot$, $5 \times 10^7 M_\odot$, $5 \times 10^8 M_\odot$

For the three different SMBH mass values (Figure 11), we see that each has different general shapes, but they all share the same characteristic of two more prominent peaks, one near $v = 0$ and one approximately at the kick velocity. The lowest mass has the most dramatic peaks with a few dimmer peaks in between. As the mass increases to $5 \times 10^7 M_\odot$, the peak at $v_{LoS} \approx 0$ becomes dimmer and is now at half the luminosity of the $v_{LoS} = 0$ peak for the $5 \times 10^6 M_\odot$ SMBH. The emission between the peaks also increases in strength. For the largest mass of $5 \times 10^8 M_\odot$, we see an overall increase in luminosity, and the peak centered near the kick velocity has become less defined and is now shaped more like a plateau. Similar to both the kick angle and kick velocity comparison, each of the features on the right side of the profile is centered near the initial kick velocity, regardless of mass.

4.3.4. *Observer Angle*

For all of the line profiles in the original simulations for Capstone II, an observer angle of 0 degrees was used. This perspective represents viewing the full disk of the AGN along its rotation axis as opposed to a side-on view. For Capstone II, I ran an additional test using a second angle for observer inclination, detailed in Figure 12. The second angle chosen is the average viewing angle for AGN, which we have derived to be 57°. This angle represents the average for a randomly oriented population of AGN. For the test itself, the two observer angles were applied to the simulation with a kick velocity of 2,000 km/s, a kick angle of 13°, and a mass of 5×10^7 Solar Masses.

The reason we see such different profile shapes for different observer angles is because of the shift in LoS velocity. To calculate the new LoS velocity when changing observer viewing angles, a rotational transformation about the x-axis adjusts the values so that the transformed z-velocity will represent the new LoS velocity. By adjusting our viewing angle, we are also adjusting how we view the cloud distribution, which in turn changes the LoS velocity relative to us, causing a very different profile shape.

5. DISCUSSION

The main goal of Capstone II was to explore the established parameter space to create a set of profile characteristics for SMBH recoils that can be used to compare with quasar databases. These properties will help us to identify likely recoil candidates from the millions of spectra in these databases.

5.1. *Profile Shapes*

From the three comparisons, we can see some clear trends in the profiles when comparing different kick velocities and SMBH masses. When comparing velocity, we see that as the magnitude of the kick increases, the number of clouds that remain with the SMBH as it moves away decreases. As the profile develops a second peak around the kick velocity, which increases in prominence relative to the peak at $v \approx 0$ because the bould clouds are more luminous, but the overall luminosity has decreased because there are fewer bound clouds close to the source of the radiation.

When comparing the kick angle, the only effect on the profile is that it determines the general shape of the cloud distribution that we see, and thus the shape of the profile. This shape is highly variable based on both the observer's viewing angle *and* the kick angle combined. There are too many combinations to consider, so for this paper, we only consider 2 observer inclinations, 0° and 57°, with a greater focus on 0°.

For the 0° observer inclination, if the kick angle is small when compared to the viewing angle (i.e., between $\approx 0° - 30°$), we see a double-peaked profile with one very bright peak near the kick velocity and one very faint peak around where $v_{LoS} = 0$. In the case of the 45° kick, we get the brightest peak around $v_{LoS} = 0$ with a tail extending to higher velocities and a broader bump around the kick velocity.

We also see a difference in luminosity between the smaller angles and 45° of an order of magnitude, with the smallest luminosity corresponding to the lower kick angles and the largest corresponding to the kick angle of 45°. The 45° kick angle has the highest luminosity because even though each of the three simulations runs for a comparable amount of time, the SMBH moving at a 45° angle will remain closer to its

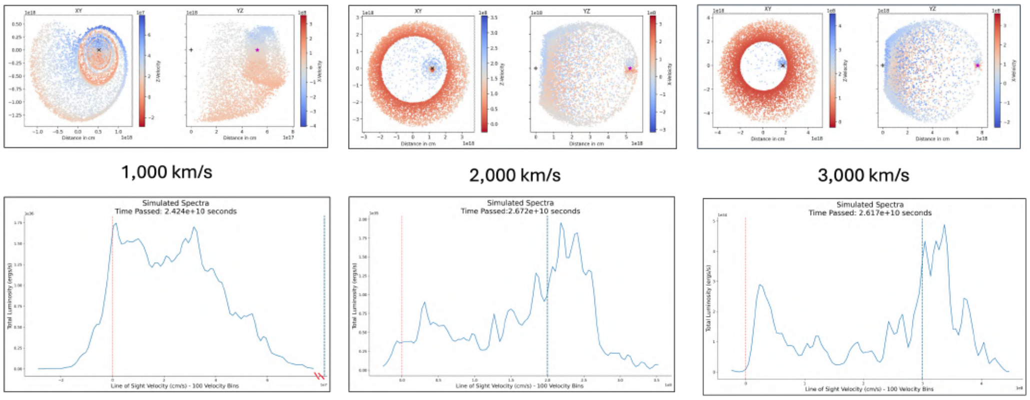


Figure 9: Comparison of line profile and spatial distribution for simulations using a 13° kick angle and a SMBH mass of $5 \times 10^7 M_\odot$ for three different kick velocities. The red line indicates $v = 0$ and the blue indicates the kick velocity in the simulation.

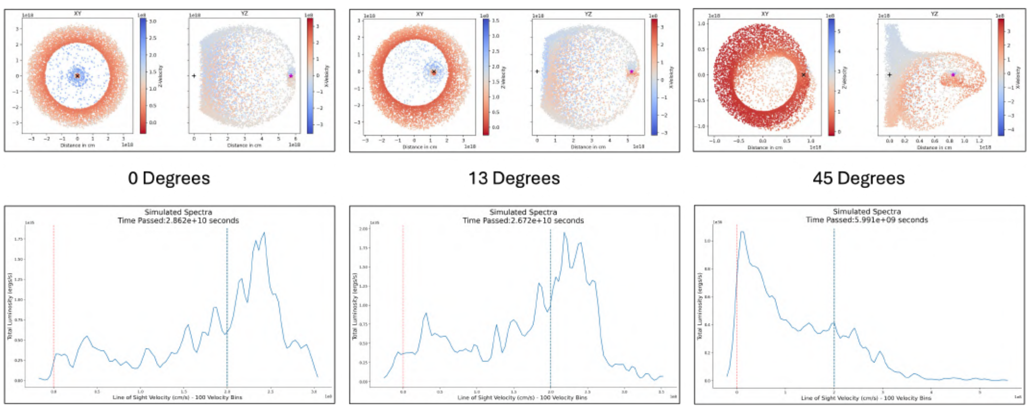


Figure 10: Comparison of the line profile and spatial distribution for simulations using a kick velocity of 2,000 km/s and a SMBH mass of $5 \times 10^7 M_\odot$ with three different kick angles. The red line indicates $v = 0$ and the blue indicates the kick velocity in the simulation.

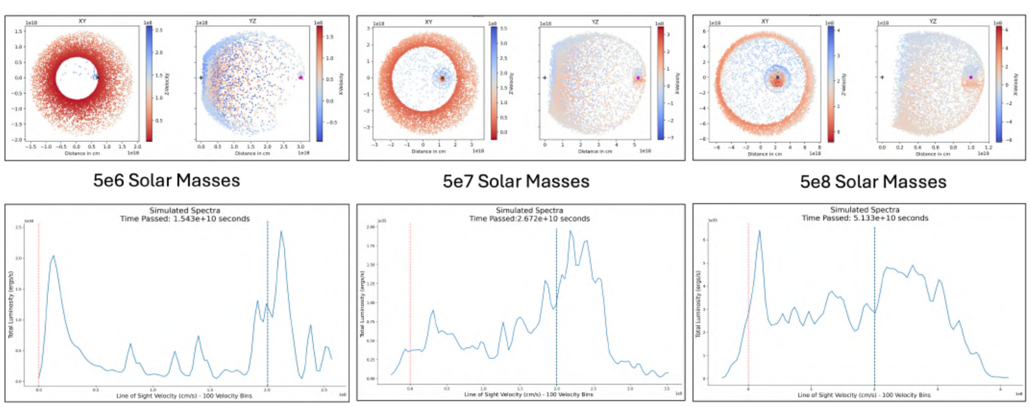


Figure 11: Comparison of the line profile and spatial distribution for a simulation using a kick velocity of 2,000 km/s and a kick angle of 45° with three different black hole masses. The red line indicates $v = 0$ and the blue indicates the kick velocity in the simulation.

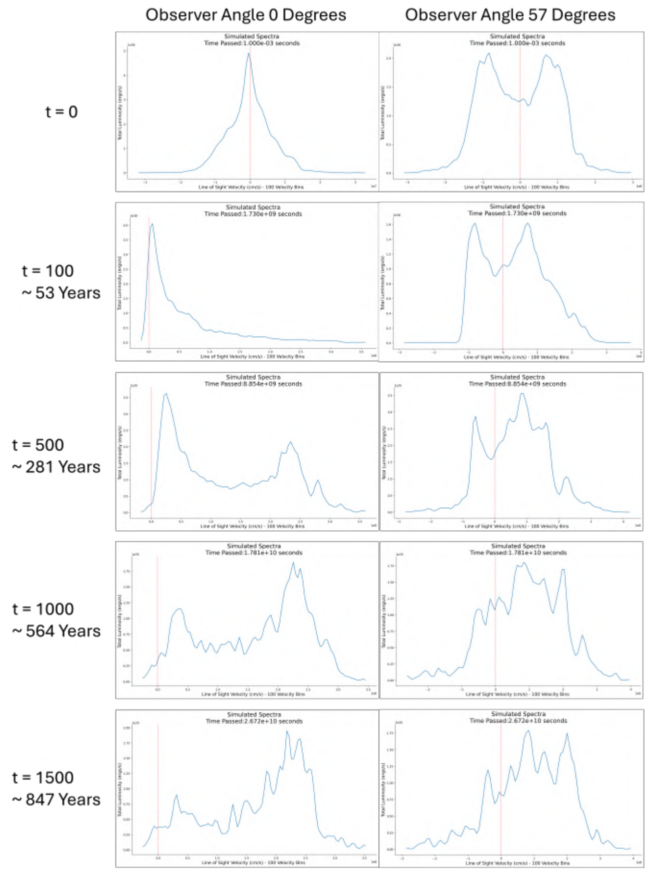


Figure 12: This figure compares a single simulation from two different observer angles. One at 0 degrees on the left, looking down at the full disk, and 57 degrees on the right, representing the average viewing angle of AGN.

starting point than it would for lower kick angles.

Lastly, for the mass comparison, we see again that the double-peaked shape for the kick angle of 13° becomes less defined as the mass increases. As mass increases, more clouds become gravitationally bound to the SMBH. We also see again that for the lowest mass SMBH, there is a difference in luminosity. With an increase in mass from $5\times10^6 M_\odot$ to $5\times10^8 M_\odot$, the luminosity of the highest peak increases by an order of magnitude. In terms of profile shape, the lowest mass has one peak at $v_{LoS} \approx 0$ and v_{kick}, while the intermediate mass has a stronger and broader peak at the kick velocity. At the highest mass, the entire profile is broader and takes on a more 'boxy' shape than the other two profiles. We believe that this is due to there still being some clouds at intermediate LoS velocities that are catching up. If we were to continue to run this simulation further, we might see the stronger peak developing at the kick velocity, like in the intermediate and low mass simulations.

To further reinforce the idea that mass and velocity have the greatest influence on profile shape, I calculated the percentage of clouds in each simulation that would theoretically follow the SMBH based on their initial Keplerian velocity (shown in Table 3). If the initial Keplerian velocity is less than the kick velocity, it is more likely to be left behind as the SMBH moves away. Based on the table, there is a much larger difference between the three simulations for velocity and mass than there is for angle, as we would expect. For the mass comparison, none of the clouds in the lowest BH mass simulations remain bound, whereas 24% of the clouds in the highest mass simulation do. The velocity comparison also shows a very dramatic difference between the lowest and highest velocities. In the case of the 1,000 km/s kick, 100% of the clouds remain bound to the SMBH, and only 0.13% remain bound for the 3,000 km/s kick.

Overall, the trends when considering the behavior of the cloud distribution and line profiles from an observer angle of 0° follow consistent patterns. For lower kick angles between 0° and 13°, a higher SMBH mass, and a lower kick velocity will result in more clouds remaining bound to the SMBH and thus produce a "brighter" and more broad profile. The particles that remain bound are fast-moving and have a higher luminosity, so they will "pull" the line profile away from the origin so a strong peak develops near where the kick velocity equals the LoS velocity. The more particles remain bound, the broader the shifted profile will be. For a kick angle of 45°, we see a different profile shape entirely, but expect to see the same trends in the velocity shift and broadening.

5.2. *Limitations of Current Model*

Identifying these profile characteristics will help in identifying possible recoils, but there are still many other things to consider. The first being that this is an imperfect model. While recombination luminosity is still a reasonable approximation, AGN-BEANS should be used as a general guide when considering the generated profiles because, in its current form, it is unable to handle the more advanced photoionization models that more accurately reflect the physics of these complex systems. This code could be adapted to handle these models in the future, but for right now, that remains out of the scope of this project. In this simulation, we also only consider the H-alpha broad lines and not the narrow lines that would also be present in the observed spectra. However, we don't expect the narrow lines to be affected because the black hole itself only moves about a parsec away from its origin during these simulations, while the narrow line region itself is hundreds of parsecs in diameter. In these simulations, the SMBH is not moving far enough away to affect the NLR in a way that we can see.

Within the n-body dynamics of the simulation itself, there is also a simplification in terms of total mass. By making the test particles massless and not allowing them to interact with each other, as well as including no other mass besides the SMBH, it greatly speeds up the runtime. This means the simulation isn't taking into account the effects of the additional mass from the surrounding galaxy. As a result, all of the particles in the simulation follow the SMBH as it moves away, and they aren't truly "left behind" like they would be if there were additional mass to hold them there. Eventually, this would become important for the dynamics of the clouds and the black hole itself. This does not greatly impact the simulation results presented here, but it means that the peak due to clouds left behind is slightly shifted away from $v_{LoS} = 0$.

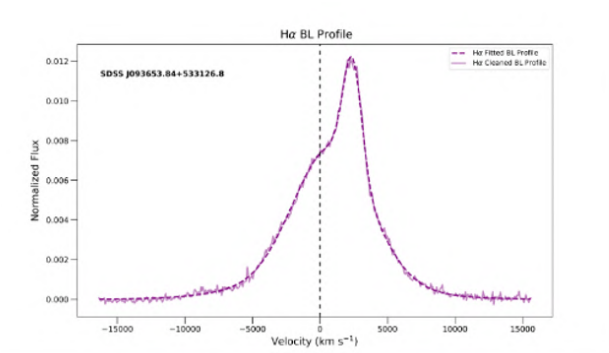
We are also assuming that the accretion disk reforms instantaneously following the kick. In reality, there may be a delay before the ionizing photon source "turns on" after the kick, but this question would have to be addressed with general relativity magnetohydrodynamic simulations.

6. APPLICATION

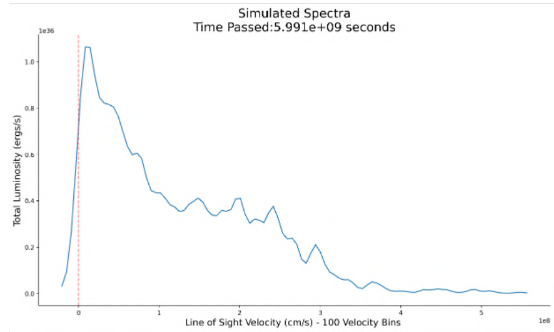
Having these simulated profiles is all well and good, but there is still one final question that has been left unanswered: Do we see broad lines in these simulations with profiles similar to line profiles in real data? Figure 13 compares two examples of broad H-alpha line profiles extracted from real AGN spectra, with profiles produced by AGN-BEANS for two different simulations. For the profile in panel (b), reversing the kick velocity would flip the profile about v_{kick}, allowing it to better line up with its counterpart in panel (a). When comparing the general shapes of the profiles, there are some strong similarities, and with further tuning, these simulations could be more closely tailored to an individual spectrum rather than comparing based solely on the overall shape.

7. CONCLUSIONS

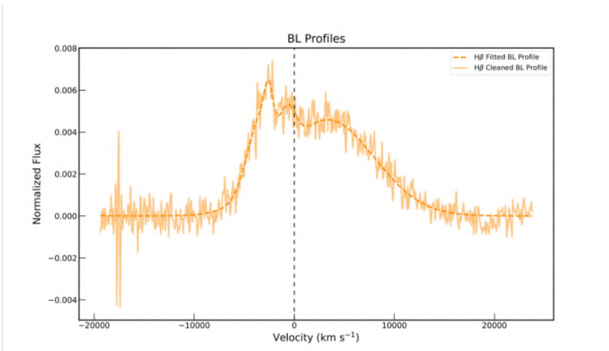
Overall, this capstone project has highlighted that the greatest effects on the emission line profiles of SMBH recoils stem from the mass of the SMBH itself and the velocity of the recoil kick. Higher kick velocities cause fewer gas clouds to remain gravitationally bound to the SMBH, and a higher SMBH mass leads to more gas clouds remaining gravitationally bound. While the kick angle does affect the cloud distribution and, therefore, the line profiles, the observed shapes also depend highly on the observer's viewing angle. As shown in Figure 12, changing the observer angle can completely change the shape



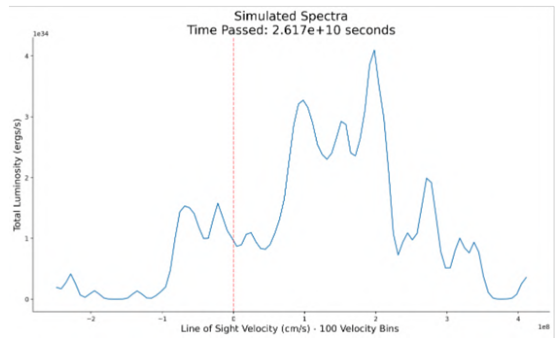
(a) SDSS 0768-52281-0473



(b) 2,000 km/s - 45° - $5 \times 10^7 M_\odot$



(c) SDSS 10742-58198-0490



(d) 3,000 km/s - 13° - $5 \times 10^7 M_\odot$

Figure 13: A comparison of two real spectra from the SDSS database, provided by Liza Matrecito, with two simulations produced by AGN-BEANS. The identification numbers for the SDSS spectra and the parameters for the simulations are included under each image.

of the profile without changing any other parameter.

By studying the changes in the profile shape over time, we also see that the intuitive prediction illustrated in Figure 2 is broadly accurate. While the change in profile shape computed by AGN-BEANS is much more complex than our original prediction, the presence of a velocity-shifted peak and broadening of the emission line are commonplace features consistent with the simple picture.

Going forward, the timescale will be the main consideration when applying these models to real data. Overall, for the simulations I have used in this paper, each recoil takes between 500-1,000 years to fully evolve to a point where the profile remains in a steady state. This means that it is unlikely we'd observe the system during the immediate aftermath of the kick while the profiles are still evolving. Our best chance would be to catch it in the steady state, long after the initial kick, where the profiles are typically broadened, shifted, and asymmetric.

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